

PUTTALAM, Sri Lanka

WMO: 434240

Lat: 8.0271N

Lon: 79.8416E

Elev: 7

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.6	70.2	64.8	91.7	77.5	66.1	96.2	76.3	11.7	83.2	9.9	82.6	1.5	50	0.330

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.9	92.6	80.0	91.6	80.3	90.7	80.2	83.2	89.2	82.5	88.8	81.9	88.1	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.5	164.1	87.6	80.8	159.7	87.0	80.1	156.1	86.4	47.2	89.5	46.4	89.1	45.7	88.5	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.7	12.0	10.8	DB	64.7	94.9	1.4	2.1	63.7	96.4	62.9	97.7	62.1	98.8	61.1	100.4	
			WB	64.0	84.1	1.4	1.2	63.0	85.0	62.1	85.7	61.3	86.4	60.3	87.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	78.5	80.4	83.1	84.1	85.0	84.4	84.1	83.8	83.6	82.4	80.5	79.2
	DBStd	2.71	1.96	2.13	1.66	1.75	1.88	1.43	1.20	1.33	1.46	1.85	1.47	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11844	884	850	1025	1023	1086	1033	1059	1049	1009	1006	915	905
	CDD65	6369	419	430	560	573	621	583	594	584	559	541	465	440
	CDH74	70327	3336	3999	6458	7128	8062	7247	7123	6861	6560	5884	4163	3506
	CDH80	26895	981	1510	2842	3254	3744	2960	2693	2494	2389	2006	1142	880
Wind	WSAvg	4.3	3.1	3.1	2.5	2.6	5.1	7.7	7.0	6.5	5.9	3.3	2.1	2.8
Precipitation	PrecAvg	46.9	1.9	1.3	2.6	6.7	3.8	1.5	0.8	0.6	2.0	9.3	9.6	6.1
	PrecMax	64.7	8.1	4.4	7.2	16.2	11.3	6.5	5.2	2.3	7.8	24.2	21.7	16.2
	PrecMin	28.8	0.0	0.0	0.0	1.5	0.0	0.0	0.0	0.0	0.0	0.1	3.6	1.1
	PrecStd	9.9	1.9	1.2	2.3	3.3	3.1	1.7	1.2	0.7	2.0	5.3	4.7	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	93.0	95.4	93.4	92.7	90.3	90.9	90.6	90.2	90.5	89.4	88.6
		MCWB	77.2	76.4	76.9	81.8	82.4	81.0	79.6	79.5	79.6	78.7	78.5	77.7
	2%	DB	87.4	90.7	92.8	92.2	91.4	89.3	89.5	89.3	88.9	89.1	88.0	87.3
		MCWB	76.3	76.8	78.2	81.8	81.9	80.6	79.7	79.4	79.4	79.2	78.8	77.4
	5%	DB	86.3	89.0	91.4	91.3	90.4	88.4	88.4	88.3	88.1	88.2	87.0	86.1
		MCWB	75.8	76.8	78.9	81.6	81.5	80.3	79.4	79.3	79.3	79.2	78.7	77.3
	10%	DB	85.1	87.5	90.2	90.3	89.4	87.6	87.5	87.3	87.2	87.1	85.7	84.7
		MCWB	75.5	76.9	79.2	81.4	81.2	80.0	79.2	79.1	79.1	79.1	78.4	76.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	80.7	82.7	84.2	84.4	82.6	81.6	81.8	81.7	81.4	80.9	80.5
		MCDB	85.9	88.0	90.3	90.4	89.6	87.9	87.7	87.7	87.6	87.4	86.6	85.8
	2%	WB	78.6	79.8	81.7	83.2	83.3	81.8	80.8	80.9	80.8	80.6	80.0	79.4
		MCDB	84.7	86.8	89.4	89.7	88.9	87.2	87.0	86.5	86.6	86.6	85.6	84.5
	5%	WB	77.7	79.0	80.9	82.6	82.5	81.2	80.1	80.1	80.1	80.0	79.4	78.6
		MCDB	83.5	85.8	88.5	89.2	88.5	86.5	86.2	85.7	85.8	85.8	84.8	83.5
	10%	WB	76.8	78.2	80.2	81.9	81.8	80.7	79.8	79.7	79.7	79.5	78.9	77.9
		MCDB	82.2	84.6	87.6	88.5	87.7	85.9	85.7	85.3	85.3	85.1	84.1	82.3
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.7	15.5	12.8	8.9	6.6	7.3	7.7	7.8	9.4	10.8	11.3
		MCDBR	15.8	17.6	17.0	13.6	10.3	7.3	8.4	8.7	8.6	10.6	12.7	13.4
		MCWBR	6.3	6.2	6.0	5.7	4.0	2.7	2.7	2.8	3.0	3.8	5.2	5.4
	5% WB	MCDBR	13.3	14.8	14.9	12.8	10.0	6.7	7.7	8.0	8.1	9.9	11.4	11.1
		MCWBR	6.1	6.4	6.3	6.1	4.5	3.0	2.9	3.1	3.4	4.2	5.2	5.3
Clear-Sky Solar Irradiance	taub	0.437	0.437	0.476	0.492	0.541	0.513	0.527	0.508	0.477	0.481	0.468	0.461	
	taud	2.351	2.325	2.236	2.222	2.069	2.161	2.104	2.169	2.272	2.266	2.300	2.298	
	Ebn at Noon	278	281	271	262	245	248	246	255	266	265	267	268	
	Edh at Noon	40	42	47	47	53	48	51	49	44	44	42	42	
All-Sky Solar Radiation	RadAvg	1611	1868	2025	1949	1819	1789	1822	1852	1877	1705	1427	1404	
	RadStd	127	103	86	87	126	83	73	83	101	92	113	126	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page